

Hybridization wave in a Kondo-Heisenberg lattice model

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Motivated by the recent experimental observation of a Kondo hybridization wave in the van der Waals material 6R-TaS₂ [1], we examine the mean-field solutions of a Kondo-Heisenberg lattice model on a triangular lattice with an enlarged unit cell. Our goal is to verify whether a hybridization wave can emerge as a self-consistent solution of a simple two-dimensional model and to gain a better understanding of the physical mechanism driving this phenomenon. Previous studies discussed how the hybridization wave can emerge near Kondo breakdown for systems whose local moments are prone to the formation a quantum spin liquid with a spinon Fermi surface [2, 3]. Since this type of spin liquid appears unfavorable in triangular-lattice systems [4–6], we explore an alternative hypothesis here, namely that the hybridization wave arises in the proximity to Kondo breakdown and the formation of a Dirac spin liquid. Results obtained for a square lattice in Ref. 7 seem compatible with this interpretation.

CONTENTS

I. Kondo-Heisenberg lattice model and its SU(M)-symmetric generalization	1
II. Large- M mean-field theory	2
A. Pure f sector: Unrestricted mean-field theory	5
1. Dimerized phase	5
2. Chiral spin liquid	6
3. Dirac spin liquid	7
B. Pure f sector: Restricted mean-field theory	7
C. c -electron Fermi surfaces	8
D. Mean-field results for the restricted theory	9
III. Progress log	9
References	10

I. KONDO-HEISENBERG LATTICE MODEL AND ITS SU(M)-SYMMETRIC GENERALIZATION

We consider a Kondo-Heisenberg lattice model

$$\mathcal{H} = - \sum_{i < j} \sum_{\alpha} \left(t_{ij} c_{i\alpha}^{\dagger} c_{j\alpha} + \text{h.c.} \right) + J_K \sum_{i, \alpha, \beta} \mathbf{S}_i \cdot \left(c_{i\alpha}^{\dagger} \frac{\boldsymbol{\sigma}_{\alpha\beta}}{2} c_{i\beta} \right) + J_H \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

defined on a single layer of a triangular lattice as a minimal model to describe the emergence of a hybridization wave. Here, $c_{i\alpha}^{\dagger}$ is an operator that creates a conduction electron with spin $\alpha = \uparrow, \downarrow$ at site i and $\mathbf{S}_i$ denotes a spin-1/2 local magnetic moment formed by strongly-interacting f electrons. We assume that the Kondo coupling J_K and the Heisenberg coupling J_H between nearest-neighbor spins are antiferromagnetic. In principle, one can work with arbitrary hopping amplitudes t_{ij} . However, for concreteness, we will restrict the c -electron hopping to nearest-neighbor sites and impose that the hopping amplitudes are real and uniform, $t_{ij} = t \in \mathbb{R}$.

To obtain a mean-field theory that allows for a controlled treatment of fluctuation effects, we introduce an SU(M)-symmetric generalization of Eq. (1):

$$\mathcal{H} = - \sum_{\langle ij \rangle, \alpha} \left(t_{ij} c_{i\alpha}^{\dagger} c_{j\alpha} + \text{h.c.} \right) + \frac{J_K}{M} \sum_{i, \alpha, \beta} S_i^{\beta\alpha} c_{i\alpha}^{\dagger} c_{i\beta} + \frac{J_H}{M} \sum_{\langle ij \rangle} \sum_{\alpha\beta} \left[S_i^{\beta\alpha} S_j^{\alpha\beta} + \frac{\kappa}{M^2} \left(S_i^{\beta\alpha} S_j^{\alpha\beta} \right)^2 \right], \quad (2)$$

where $\alpha = 1, \dots, M$ and

$$S_i^{\beta\alpha} = f_{i\beta}^{\dagger} f_{i\alpha} - \frac{Q}{M} \delta_{\alpha\beta} \quad (3)$$

are $SU(M)$ generators subject to the constraint

$$\sum_{\alpha=1}^M f_{i\alpha}^\dagger f_{i\alpha} = Q. \quad (4)$$

The value of the charge Q defines the representation of $SU(M)$ under consideration. Here, we fix $Q = M/2$, which corresponds to the totally antisymmetric irreducible representation of $SU(M)$ and connects to the usual Abrikosov representation in the case $M = 2$. Note that Eq. (1) includes an additional biquadratic interaction whose strength is parametrized by the dimensionless quantity κ . We have introduced this term to suppress the strong dimerization tendencies that are favored at large M [7, 8], but are not expected to be relevant for the physical $M = 2$ case.

II. LARGE- M MEAN-FIELD THEORY

The first step required to develop a mean-field theory is to fix a unit cell. In lieu of keeping the single-site crystallographic unit cell of the triangular lattice and allowing the mean-field parameters to vary with an unspecified ordering wavevector $\mathbf{Q}$, we assume a two-site unit cell, as illustrated in Fig. 1. Note that this choice is compatible with the hybridization wave measured in Ref. 1. The primitive vectors of the direct and reciprocal lattices are then given by

$$\begin{cases} \mathbf{a}_1 = (1, 0) \\ \mathbf{a}_2 = (1, \sqrt{3}) \end{cases} \quad \Rightarrow \quad \begin{cases} \mathbf{b}_1 = 2\pi(1, -1/\sqrt{3}), \\ \mathbf{b}_2 = 2\pi(0, 1/\sqrt{3}), \end{cases} \quad (5)$$

if we set the lattice spacing to one. For future convenience, we define the nearest-neighbor vectors $\boldsymbol{\delta}_1 = (1, 0)$, $\boldsymbol{\delta}_2 = (1, \sqrt{3})/2$, and $\boldsymbol{\delta}_3 = (-1, \sqrt{3})/2$. It will also prove useful to label sites in terms of a composite index (ia) , where i numbers one of the N_c and $a \in \{A \equiv 1, B \equiv 2\}$ is a sublattice index. We shall denote the total number of sites as $N = 2N_c$.

With the previous conventions, the c -electron hopping term can be written as

$$\begin{aligned} \mathcal{H}_t &= -t \sum_{ia} \sum_{n=1}^3 c_{(ia),\alpha}^\dagger [c_{(ia)+\boldsymbol{\delta}_n,\alpha} + c_{(ia)-\boldsymbol{\delta}_n,\alpha}] \\ &= -2t \sum_{\mathbf{k}} \left[\cos k_1 \sum_a c_{\mathbf{k}a\alpha}^\dagger c_{\mathbf{k}a\alpha} + \left(\sum_{n=2}^3 \cos k_n \right) (c_{\mathbf{k}A\alpha}^\dagger c_{\mathbf{k}B\alpha} + \text{h.c.}) \right], \end{aligned} \quad (6)$$

where $k_n = \mathbf{k} \cdot \boldsymbol{\delta}_n$ and sums over repeated flavor (spin) indices are implied. To apply the correct large- M decoupling to the Kondo and Heisenberg terms, we use the identities

$$\begin{aligned} \mathcal{H}_K &= \frac{J_K}{M} \sum_{ia} \left(f_{ia\beta}^\dagger f_{ia\alpha} - \frac{1}{2} \delta_{\alpha\beta} \right) c_{ia\alpha}^\dagger c_{ia\beta} \\ &= \frac{J_K}{M} \sum_{ia} \left(\frac{M}{2} - \frac{1}{2} c_{ia\alpha}^\dagger c_{ia\alpha} - f_{ia\beta}^\dagger c_{ia\beta} c_{ia\alpha}^\dagger f_{ia\alpha} \right) \end{aligned} \quad (7)$$

and

$$\begin{aligned} \mathcal{H}_H &= \frac{J_H}{M} \sum_{\langle ia,jb \rangle} \left(f_{ia\beta}^\dagger f_{ia\alpha} - \frac{1}{2} \delta_{\alpha\beta} \right) \left(f_{jb\alpha}^\dagger f_{jb\beta} - \frac{1}{2} \delta_{\alpha\beta} \right) \\ &= \frac{J_H}{M} \sum_{\langle ia,jb \rangle} \left(\frac{M}{4} - f_{ia\beta}^\dagger f_{jb\beta} f_{jb\alpha}^\dagger f_{ia\alpha} \right). \end{aligned} \quad (8)$$

Defining the mean fields

$$b_a = \frac{1}{M} \sum_{\sigma} \langle f_{ia\sigma}^\dagger f_{ia\sigma} \rangle, \quad \chi_{jb,ia} = \frac{1}{M} \sum_{\sigma} \langle f_{jb\sigma}^\dagger f_{ia\sigma} \rangle = \chi_{ia,jb}^*, \quad (9)$$

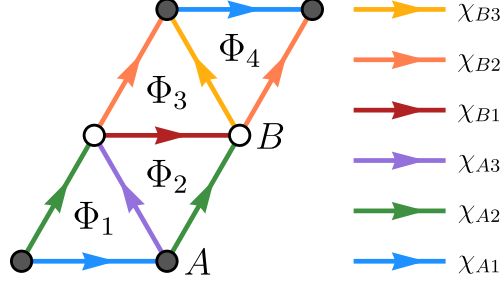


FIG. 1. Mean-field unit cell, depicting the two sublattices, A and B, the convention for the direction of the χ fields, and the four independent fluxes, $\Phi_1, \dots, \Phi_4$.

and dropping irrelevant constants, we obtain

$$\mathcal{H}_K \approx -\frac{J_K}{2M} \sum_{ia} c_{ia\sigma}^\dagger c_{ia\sigma} - J_K \sum_{ia} \left(b_a^* c_{ia\sigma}^\dagger f_{ia\sigma} + \text{h.c.} - M|b_a|^2 \right), \quad (10)$$

$$\mathcal{H}_H \approx -J_H \sum_{\langle ia,jb \rangle} \left(\chi_{jb,ia} f_{ia\alpha}^\dagger f_{jb\alpha} + \text{h.c.} - M|\chi_{jb,ia}|^2 \right). \quad (11)$$

To treat the biquadratic interaction, we square the decoupling of $S_i^{\beta\alpha} S_j^{\alpha\beta}$ and apply the same large- M decoupling to the remaining quartic contribution [7]. The result reads

$$\begin{aligned} \frac{1}{M^2} \left(S_i^{\beta\alpha} S_j^{\alpha\beta} \right)^2 &\approx \left(M|\chi_{ji}|^2 - \chi_{ji} f_{i\alpha}^\dagger f_{j\alpha} - \chi_{ij} f_{j\alpha}^\dagger f_{i\alpha} \right)^2 \\ &\approx -3M^2 |\chi_{ji}|^4 + 2M |\chi_{ji}|^2 \left(\chi_{ji} f_{i\alpha}^\dagger f_{j\alpha} + \chi_{ji}^* f_{j\alpha}^\dagger f_{i\alpha} \right). \end{aligned} \quad (12)$$

Finally, we introduce static, sublattice-dependent Lagrange multipliers λ_a to enforce that the constraints in Eq. (4) are satisfied *on average* and, with this, obtain the mean-field Hamiltonian

$$\begin{aligned} \mathcal{H}_{\text{MF}} = \mathcal{H}_t &+ \sum_{ia} \left[\lambda_a \left(f_{ia\sigma}^\dagger f_{ia\sigma} - \frac{M}{2} \right) - \frac{J_K}{2M} c_{ia\sigma}^\dagger c_{ia\sigma} \right] - J_K \sum_{ia} \left(b_a^* c_{ia\sigma}^\dagger f_{ia\sigma} + \text{h.c.} - M|b_a|^2 \right) \\ &- J_H \sum_{\langle ia,jb \rangle} \left[\tilde{\chi}_{jb,ia} f_{ia\alpha}^\dagger f_{jb\alpha} + \text{h.c.} + M|\chi_{jb,ia}|^2 \left(3\kappa |\chi_{jb,ia}|^2 - 1 \right) \right], \end{aligned} \quad (13)$$

where

$$\tilde{\chi}_{jb,ia} = \left(1 - 2\kappa |\chi_{jb,ia}|^2 \right) \chi_{jb,ia}. \quad (14)$$

Given that the unit cell depicted in Fig. 1 accommodates six nearest-neighbor bonds, a translationally-invariant mean-field solution generally has six independent complex fields $\chi_{an} \equiv \chi_{(ia)+\delta_n, (ia)}$. Two of these, χ_{A1} and χ_{B1} , connect sites on the same sublattice.

After performing a Fourier transform, we can write the grand-canonical operator $\mathcal{K}_{\text{MF}} = \mathcal{H}_{\text{MF}} - \mu \mathcal{N}_c$, where $\mathcal{N}_c = \sum_{ia\sigma} c_{ia\sigma}^\dagger c_{ia\sigma}$ is the total number operator for the c electrons, as

$$\mathcal{K}_{\text{MF}} = E_0 + \sum_{\mathbf{k}\sigma} \Psi_{\mathbf{k}\sigma}^\dagger \underbrace{\begin{pmatrix} tG(\mathbf{k}) - \mu \mathbb{1}_2 & -J_K B^* \\ -J_K B & J_H G_\chi(\mathbf{k}) + \Lambda \end{pmatrix}}_{=\hbar_{\mathbf{k}}} \Psi_{\mathbf{k}\sigma}, \quad (15)$$

with $\Psi_{\mathbf{k}\sigma}^\top = (c_{\mathbf{k}A\sigma} \ c_{\mathbf{k}B\sigma} \ f_{\mathbf{k}A\sigma} \ f_{\mathbf{k}B\sigma})$ and

$$E_0 = MN_c \left[\sum_a \left(J_K |b_a|^2 - \frac{\lambda_a}{2} \right) + J_H \sum_{an} |\chi_{an}|^2 \left(1 - 3\kappa |\chi_{an}|^2 \right) \right]. \quad (16)$$

The 4×4 matrix $h_{\mathbf{k}}$ is defined in terms of the 2×2 submatrices $B_{ab} = b_a \delta_{ab}$, $\Lambda_{ab} = \lambda_a \delta_{ab}$, and

$$G(\mathbf{k}) = -2 \begin{pmatrix} \cos k_1 & \cos k_2 + \cos k_3 \\ \cos k_2 + \cos k_3 & \cos k_1 \end{pmatrix}, \quad (17)$$

$$G_\chi(\mathbf{k}) = - \begin{pmatrix} 2 \operatorname{Re}(e^{ik_1} \tilde{\chi}_{A1}) & \sum_{n=2}^3 (e^{ik_n} \tilde{\chi}_{An} + e^{-ik_n} \tilde{\chi}_{Bn}^*) \\ \sum_{n=2}^3 (e^{ik_n} \tilde{\chi}_{Bn} + e^{-ik_n} \tilde{\chi}_{An}^*) & 2 \operatorname{Re}(e^{ik_1} \tilde{\chi}_{B1}) \end{pmatrix}. \quad (18)$$

Note that the J_K -dependent on-site potential in Eq. (13) has been absorbed into the definition of μ . Changes in the c -electron hopping amplitudes can be accounted for by modifying $G(\mathbf{k})$ provided that they do not enlarge the unit cell. While the fields b_a are in principle complex, we can eliminate their imaginary parts by performing a sublattice-dependent gauge transformation $c_{\mathbf{k}a\sigma} \mapsto e^{-i\varphi_a} c_{\mathbf{k}a\sigma}$ with $\varphi_a = \arg(b_a)$. Unless stated otherwise, we will work in the gauge where all b fields are real.

As usual, the quadratic operator in Eq. (15) can be diagonalized via $\mathbf{k}$ -dependent unitary transformations

$$\mathcal{K}_{\text{MF}} = E_0 + \sum_{\mathbf{k}\sigma} \Psi_{\mathbf{k}\sigma}^\dagger U_{\mathbf{k}} U_{\mathbf{k}}^\dagger h_{\mathbf{k}} U_{\mathbf{k}} U_{\mathbf{k}}^\dagger \Psi_{\mathbf{k}\sigma} \equiv E_0 + \sum_{\mathbf{k}\sigma} \sum_{n=1}^4 E_{\mathbf{k}n} \phi_{\mathbf{k}n\sigma}^\dagger \phi_{\mathbf{k}n\sigma}, \quad (19)$$

where $E_{\mathbf{k}n}$ are the eigenvalues of $h_{\mathbf{k}}$, $U_{\mathbf{k}}$ is the unitary matrix whose columns correspond to the eigenvectors of $h_{\mathbf{k}}$. The original c and f fermionic operators are related to the new ϕ operators via the basis transformation

$$\Psi_{\mathbf{k}\sigma} = U_{\mathbf{k}} \Phi_{\mathbf{k}\sigma} \quad \Longleftrightarrow \quad \Psi_{\mathbf{k}\sigma,m} = U_{\mathbf{k}}^{mn} \phi_{\mathbf{k}n\sigma}. \quad (20)$$

If we fix the c -electron filling $n_c = \langle \mathcal{N}_c \rangle / N$, where $N = N_s N_c = 2N_c$ denotes the total number of sites, then we have a total of eight self-consistency equations:

$$n_c = \frac{1}{N} \sum_{\mathbf{k}a\sigma} \langle c_{\mathbf{k}a\sigma}^\dagger c_{\mathbf{k}a\sigma} \rangle \quad \Longrightarrow \quad n_c = \frac{M}{N} \sum_{\mathbf{k}} \sum_{a=1}^{N_s} \sum_{n=1}^{2N_s} |U_{\mathbf{k}}^{an}|^2 n_F(E_{\mathbf{k}n}), \quad (21a)$$

$$\frac{M}{2} = \frac{1}{N_c} \sum_{\mathbf{k}\sigma} \langle f_{\mathbf{k}a\sigma}^\dagger f_{\mathbf{k}a\sigma} \rangle \quad \Longrightarrow \quad \frac{1}{2} = \frac{1}{N_c} \sum_{\mathbf{k}} \sum_{n=1}^{2N_s} |U_{\mathbf{k}}^{N_s+a,n}|^2 n_F(E_{\mathbf{k}n}), \quad (21b)$$

$$b_a = \frac{1}{MN_c} \sum_{\mathbf{k}\sigma} \langle c_{\mathbf{k}a\sigma}^\dagger f_{\mathbf{k}a\sigma} \rangle \quad \Longrightarrow \quad b_a = \frac{1}{N_c} \sum_{\mathbf{k}} \sum_{n=1}^{2N_s} (U_{\mathbf{k}}^{an})^* U_{\mathbf{k}}^{N_s+a,n} n_F(E_{\mathbf{k}n}), \quad (21c)$$

$$\chi_{a1} = \frac{1}{MN_c} \sum_{\mathbf{k}\sigma} e^{-ik_1} \langle f_{\mathbf{k}a\sigma}^\dagger f_{\mathbf{k}a\sigma} \rangle \quad \Longrightarrow \quad \chi_{a1} = \frac{1}{N_c} \sum_{\mathbf{k}} \sum_{n=1}^{2N_s} e^{-ik_1} |U_{\mathbf{k}}^{N_s+a,n}|^2 n_F(E_{\mathbf{k}n}), \quad (21d)$$

$$\chi_{a\ell} = \frac{1}{MN_c} \sum_{\mathbf{k}\sigma} e^{-ik_\ell} \langle f_{\mathbf{k}\bar{a}\sigma}^\dagger f_{\mathbf{k}a\sigma} \rangle \quad \Longrightarrow \quad \chi_{a\ell} = \frac{1}{N_c} \sum_{\mathbf{k}} \sum_{n=1}^{2N_s} e^{-ik_\ell} (U_{\mathbf{k}}^{N_s+\bar{a},n})^* U_{\mathbf{k}}^{N_s+a,n} n_F(E_{\mathbf{k}n}), \quad (21e)$$

where $n_F(x) = (e^{\beta x} + 1)^{-1}$ is the Fermi-Dirac distribution function at inverse temperature $\beta = T^{-1}$. In Eq. (21e), $\bar{a}$ represents the sublattice opposite to a , i.e., $\bar{a} = B$ if $a = A$ and vice-versa, and $\ell = 2, 3$.

While the distinction between two mean-field phases may arise from differences in the hybridization fields b_a , it may also be expressed in gauge-invariant quantities obtained from the spinon hopping fields. These quantities consist of the absolute values $|\chi_{\mu n}|$ of the bond variables and the four inequivalent fluxes

$$\begin{aligned} \Phi_1 &= \arg(\chi_{A2} \chi_{A3}^* \chi_{A1}^*), & \Phi_3 &= \arg(\chi_{B2} \chi_{B3}^* \chi_{B1}^*), \\ \Phi_2 &= \arg(\chi_{B1} \chi_{A2}^* \chi_{A3}), & \Phi_4 &= \arg(\chi_{A1} \chi_{B2}^* \chi_{B3}), \end{aligned} \quad (22)$$

depicted in Fig. 1. Note that the fluxes are defined in a clockwise orientation for every plaquette.

Finally, in order to be able to discern the ground states among multiple coexisting mean-field solutions, we must compute the (Helmholtz) free energy:

$$\begin{aligned} F &= -T \ln(\operatorname{Tr} e^{-\beta \mathcal{H}_{\text{MF}}}) = -T \ln(\operatorname{Tr} e^{-\beta \mathcal{K}_{\text{MF}}}) + N\mu n_c \\ &= E_0 + N\mu n_c - TM \sum_{\mathbf{k}} \sum_{n=1}^{2N_s} \ln(1 + e^{-\beta E_{\mathbf{k}n}}). \end{aligned} \quad (23)$$

A. Pure f sector: Unrestricted mean-field theory

When J_K/J_H is sufficiently small, we expect Kondo screening to break down in the ground state of the mean-field theory. As a result, the f spinons decouple from the conduction electrons and, within the large- M theory, form some kind of quantum disordered phase. This can either be a quantum spin liquid or a dimerized phase depending on the geometry of the lattice and the value of κ . Our goal now is to derive the optimal state in the f sector as a function of κ for the two-site unit cell in Fig. 1. To this end, we split the mean-field Hamiltonian into c and f blocks, $\mathcal{H}_{\text{MF}} = \mathcal{H}_{\text{MF}}^c + \mathcal{H}_{\text{MF}}^f$, and focus on the latter. We then diagonalize

$$\mathcal{H}_{\text{MF}}^f = E_0^f + \sum_{\mathbf{k}\sigma} \begin{pmatrix} f_{\mathbf{k}A\sigma}^\dagger & f_{\mathbf{k}B\sigma}^\dagger \end{pmatrix} \begin{pmatrix} \lambda_A - 2J_H \text{Re}(e^{ik_1} \tilde{\chi}_{A1}) & -J_H \sum_{n=2}^3 (e^{ik_n} \tilde{\chi}_{An} + e^{-ik_n} \tilde{\chi}_{Bn}^*) \\ -J_H \sum_{n=2}^3 (e^{ik_n} \tilde{\chi}_{Bn} + e^{-ik_n} \tilde{\chi}_{An}^*) & \lambda_B - 2J_H \text{Re}(e^{ik_1} \tilde{\chi}_{B1}) \end{pmatrix} \begin{pmatrix} f_{\mathbf{k}A\sigma} \\ f_{\mathbf{k}B\sigma} \end{pmatrix}, \quad (24)$$

to obtain the spectrum

$$\epsilon_{\mathbf{k}\pm}^f = \frac{(\lambda_A + \lambda_B)}{2} - J_H \text{Re}[(\tilde{\chi}_{A1} + \tilde{\chi}_{B1})e^{ik_1}] \pm J_H \sqrt{\left\{ \frac{(\lambda_A - \lambda_B)}{2J_H} - \text{Re}[(\tilde{\chi}_{A1} - \tilde{\chi}_{B1})e^{ik_1}] \right\}^2 + \left| \sum_{n=2}^3 (e^{ik_n} \tilde{\chi}_{An} + e^{-ik_n} \tilde{\chi}_{Bn}^*) \right|^2}. \quad (25)$$

After conducting an unrestricted search for self-consistent solutions of Eqs. (21b), (21d), and (21e), we determined that the mean-field ground state within the pure f sector exhibits strong dimerization for $\kappa = 0$. This is in agreement with a trend observed for the square lattice [8, 9]. While a negative κ appeared to increase the stability of the dimerized phase, we found that this ground state is eventually destabilized with increasing $\kappa > 0$ (see Fig. 2). Above a critical $\kappa_c \approx 0.3$, the system enters a gapped chiral spin liquid with a $\pi/2$ flux on every triangular plaquette. In the following, we provide further analytical understanding of these two phases.

1. Dimerized phase

A typical configuration for the dimerized (VBS) phase that arises near $\kappa = 0$ is depicted in Fig. 2. From the numerical solution of the mean-field equations, we find that this phase is characterized by parameters

$$\begin{aligned} \lambda_A = \lambda_B = 0, & \quad \chi_{A1} = -\chi_{B1} = \chi_1 e^{i\phi_1}, \\ \chi_{A2} = \chi_2 e^{i\phi_2}, & \quad \chi_{B2} = \chi_{A3} = \chi_{B3} = 0, \end{aligned} \quad (26)$$

with $0 < \chi_1 \ll \chi_2$. We observe that the strength of the dimerization decreases with increasing κ . Substituting the ansatz in Eq. (26) into Eq. (25), we find

$$\epsilon_{\mathbf{k}\pm}^f = \pm J_H \sqrt{4\tilde{\chi}_1^2 \cos^2(k_1 + \phi_1) + \tilde{\chi}_2^2}. \quad (27)$$

This shows that the bands are essentially flat in the limit $\chi_1/\chi_2 \rightarrow 0$, as one would expect for a strongly dimerized phase. The gap between the lower and upper flat bands is then given by $\Delta \approx J_H$. As a side note, let us mention that our numerical results yield various different values of $\lambda \neq 0$ depending on the initial configuration. We rationalize this by noting that, as long as the distances between λ and the flat bands is large compared to T , small variations in λ give rise to exponentially small changes in the mean-field equations. Hence, the value of λ is not fixed in a numerical implementation with limited accuracy. However, we can infer that $\lambda = 0$ by the particle-hole symmetry of the bands.

Let us now compute the energy per site and flavor analytically for a perfectly dimerized phase. To this end, we set $\chi_1 = 0$, $\chi_2 = \chi$, and $\phi_2 = \phi$. Then, taking the limit $T \rightarrow 0$, we find that

$$\begin{aligned} f_0(\kappa, \chi) &\equiv \frac{1}{MN} \lim_{T \rightarrow 0} F_f(T, \kappa, \chi) = \frac{E_0^f}{MN} + \frac{1}{N} \sum_{\mathbf{k}} \epsilon_{\mathbf{k}-}^f \\ &= \frac{J_H}{2} [\chi^2(1 - 3\kappa\chi^2) - |\chi(1 - 2\kappa\chi^2)|] \\ &= \frac{J_H}{2} (-3\kappa\chi^4 \pm 2\kappa\chi^3 + \chi^2 \mp \chi), \end{aligned} \quad (28)$$

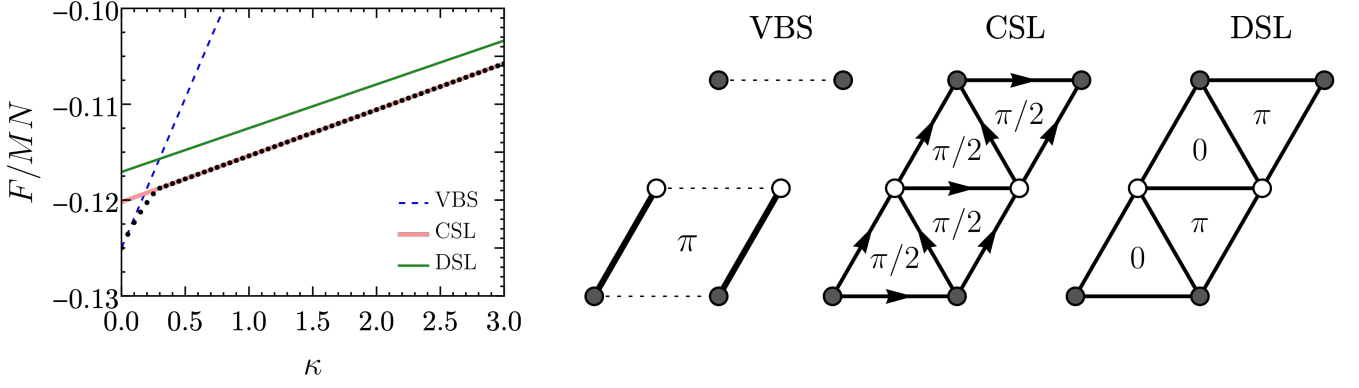


FIG. 2. Left: Free energy per site and flavor as a function of the biquadratic coupling κ . The black dots represent numerical data, whereas the colored lines correspond to analytic expressions obtained from different ansätze considered in the main text. Right: Sketch of the unit cells of the different variational states considered as possible solutions of the pure f sector.

where the $\pm$ and $\mp$ signs in front of the cubic and linear terms come from the absolute value. Since f_0 is unbounded from below when $\kappa > 0$, the system does not have a well-defined global minimum. From

$$\frac{\partial f_0}{\partial \chi} = \frac{J_H}{2} (2\chi \mp 1)(1 - 6\kappa\chi^2), \quad \frac{\partial^2 f_0}{\partial \chi^2} = \frac{J_H}{2} (-36\kappa\chi^2 \pm 12\kappa\chi + 2), \quad (29)$$

we see that f_0 has two extrema at $\chi' = \pm 1/2$ and $\chi''^2 = 1/(6\kappa)$. The second derivatives at these points are

$$\frac{\partial^2 f_0}{\partial \chi^2}(\chi') = \frac{3J_H}{2} \left(\frac{2}{3} - \kappa \right), \quad \frac{\partial^2 f_0}{\partial \chi^2}(\chi'') = 2J_H \left(\sqrt{\frac{3\kappa}{2}} - 1 \right). \quad (30)$$

Thus, χ' and χ'' are respectively a minimum and a maximum for $\kappa < 2/3$, but they “collide” at $\kappa = 2/3$ and switch roles once $\kappa > 2/3$. The energy of the local minimum is then given by

$$f_{\text{VBS}}(\kappa) = \begin{cases} \frac{J_H}{8} \left(\frac{\kappa}{4} - 1 \right) & \text{for } \kappa \leq 2/3, \\ \frac{J_H}{6} \left(\frac{1}{4\kappa} - \sqrt{\frac{2}{3\kappa}} \right) & \text{for } \kappa > 2/3, \end{cases} \quad (31)$$

which is plotted as a dashed blue line in Fig. 2. Note that this yields the correct result $f_0 = -0.125J_H$ for $\kappa = 0$, but deviates from the numerical solutions for any $\kappa > 0$. This deviation is due to an oversimplified assumption that $\chi_1 = 0$, which does not hold for $\kappa > 0$.

2. Chiral spin liquid

By inspecting our numerical results, we find that the ground state at large κ is characterized by χ fields with the same absolute value and that yield a $\pi/2$ flux on every elementary triangular plaquette (see Fig. 2). This state breaks time-reversal symmetry, and is therefore a chiral spin liquid. Imposing the ansatz

$$\begin{aligned} \lambda_A = \lambda_B = 0, & & \chi_{A2} = \chi_{B2} = +\chi, \\ \chi_{A1} = -\chi_{B1} = i\chi, & & \chi_{A3} = -\chi_{B3} = -\chi, \end{aligned} \quad (32)$$

we find that the spectrum of this phase is given by

$$\epsilon_{\mathbf{k}\pm}^f = \pm 2J_H |\chi(1 - 2\kappa\chi^2)| \sqrt{\sin^2 k_1 + \cos^2 k_2 + \sin^2 k_3} \quad (33)$$

and, in the zero-temperature limit, its free energy per flavor and site reads

$$\begin{aligned} f_0(\kappa, \chi) &\equiv 3\chi^2(1 - 3\kappa\chi^2) - \alpha |\chi(1 - 2\kappa\chi^2)| \\ &= J_H(-9\kappa\chi^4 \pm 2\alpha\kappa\chi^3 + 3\chi^2 \mp \alpha\chi), \end{aligned} \quad (34)$$

with

$$\begin{aligned}\alpha &= \frac{1}{N_c} \sum_{\mathbf{k}} \sqrt{\sin^2 k_1 + \cos^2 k_2 + \sin^2 k_3} \\ &\mapsto \frac{\sqrt{3}}{4\pi^2} \int_{\text{BZ}} d^2k \sqrt{\sin^2 k_1 + \cos^2 k_2 + \sin^2 k_3} \approx 1.20102.\end{aligned}\quad (35)$$

One can easily show that f_0 has stationary points at $\chi' = \pm\alpha/6$ and $\chi'' = 1/\sqrt{6\kappa}$. Similarly to the case of the dimerized phase, χ' starts out as a local minimum for small κ , but turns into a maximum once it collides with χ'' at $\kappa = 6/\alpha^2 \approx 4.16$. Thus, the optimal energy obtained from this ansatz is

$$f_{\text{CSL}}(\kappa) = \begin{cases} \frac{J_H \alpha^2}{12} \left(\frac{\alpha^2}{36} \kappa - 1 \right) & \text{for } \kappa \leq 6/\alpha^2, \\ J_H \left(\frac{1}{4\kappa} - \frac{\alpha}{9} \sqrt{\frac{6}{\kappa}} \right) & \text{for } \kappa > 6/\alpha^2, \end{cases} \quad (36)$$

This function is plotted as a red line in Fig. 2 and matches the numerical data perfectly for the values of κ between $\kappa_c \approx 0.30$ and the maximum κ considered in the plot.

3. Dirac spin liquid

For completeness, we also consider the ansatz

$$\lambda_A = \lambda_B = 0, \quad \chi_{A\pi} = \chi_{B2} = +\chi, \quad \chi_{B1} = \chi_{B3} = -\chi, \quad (37)$$

with $\chi \in \mathbb{R}$. One can verify that this yields a Dirac spin liquid, in which up and down triangular plaquettes are threaded by 0 and π fluxes, respectively, when $\chi > 0$. The dispersion and energy per site and flavor of this state are given by

$$\epsilon_{\mathbf{k}\pm}^f = \pm 2J_H |\chi(1 - 2\kappa\chi^2)| \sqrt{\cos^2 k_1 + \cos^2 k_2 + \sin^2 k_3}, \quad (38)$$

$$f_0 = 3\chi^2(1 - 3\kappa\chi^2) - \gamma |\chi(1 - 2\kappa\chi^2)|, \quad (39)$$

with

$$\gamma = \frac{1}{N_c} \sum_{\mathbf{k}} \sqrt{\cos^2 k_1 + \cos^2 k_2 + \sin^2 k_3} \approx 1.18524. \quad (40)$$

One can show that the Dirac points of the dispersion (38) are located at $\mathbf{K}_{\pm} = \pm(\frac{\pi}{2}, \frac{\pi}{2\sqrt{3}})$ and momenta that are equivalent up to translations by reciprocal lattice vectors.

The variational energy obtained from the ansatz (37) is identical to Eq. (36) apart from the replacement $\alpha \rightarrow \gamma$:

$$f_{\text{DSL}}(\kappa) = \begin{cases} \frac{J_H \gamma^2}{12} \left(\frac{\gamma^2}{36} \kappa - 1 \right) & \text{for } \kappa \leq 6/\gamma^2, \\ J_H \left(\frac{1}{4\kappa} - \frac{\gamma}{9} \sqrt{\frac{6}{\kappa}} \right) & \text{for } \kappa > 6/\gamma^2. \end{cases} \quad (41)$$

The fact that $\gamma < \alpha$ determines that the Dirac spin liquid is less favorable than the chiral spin liquid above.

B. Pure f sector: Restricted mean-field theory

The previous subsection showed that a Dirac spin liquid does not naturally emerge as a ground state of our full large- M mean field theory. To obtain this state as a self-consistent solution, we can restrict ourselves to real χ variables, and thus eliminate the chiral spin liquid solution. For small κ , we still obtained a dimerized phase; however, as shown in Fig. 3, the Dirac spin liquid described in Sec. II A 3 becomes the ground state for sufficiently large κ . The spectrum of this state and its free energy per site and flavor are given exactly by Eqs. (38) and (41).

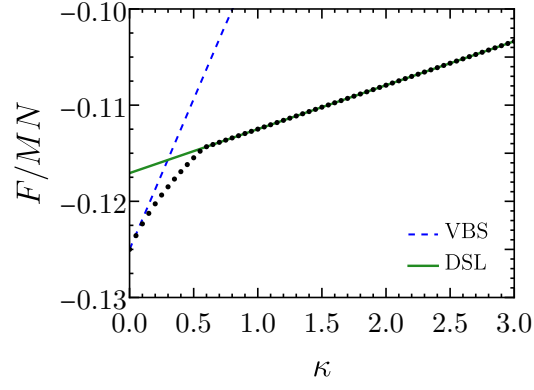


FIG. 3. Free energy per site and flavor as a function of the biquadratic coupling κ for a theory with real bond variables χ . The black dots represent numerical data, whereas the colored lines correspond to analytic expressions obtained from different ansätze considered in the main text.

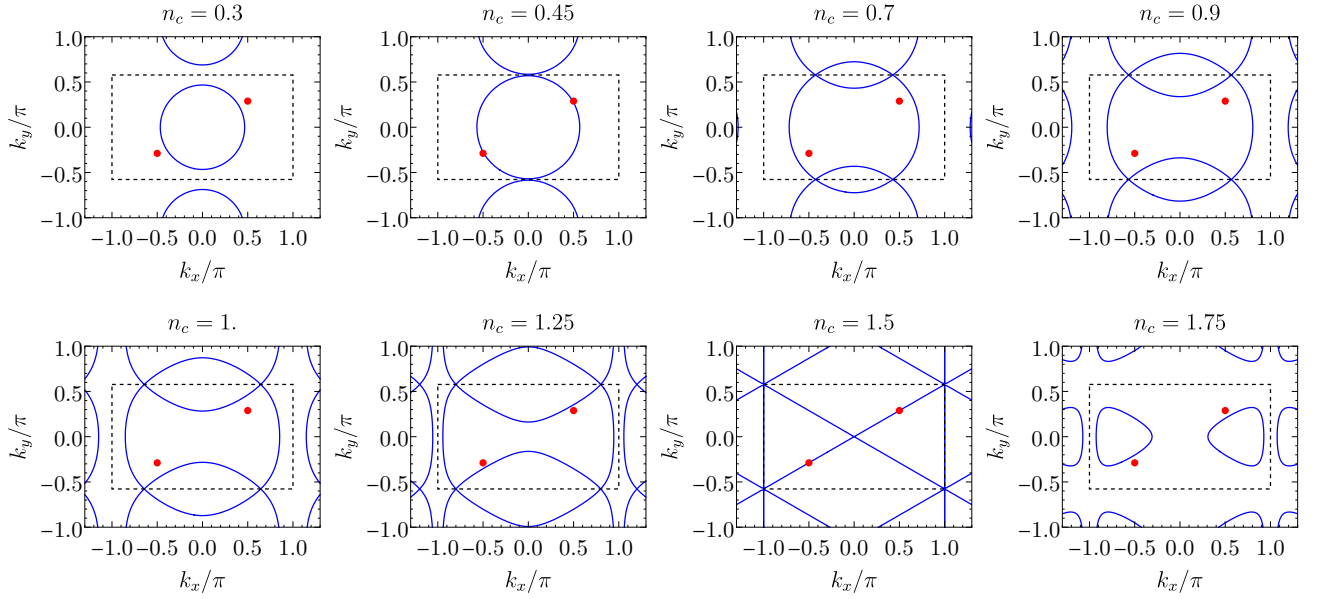


FIG. 4. Bare c Fermi surfaces for different fillings. The Brillouin zone associated with the two-site mean-field unit cell is shown in dashed lines, and the momenta $\mathbf{K}_{\pm} = \pm(\frac{\pi}{2}, \frac{\pi}{\sqrt{3}})$ corresponding to the locations of the Dirac cones of the Dirac spin liquid are highlighted in red circles. Note that there are two Lifshitz transitions as a function of n_c .

C. c -electron Fermi surfaces

To gain insight into the ways in which the lattice Kondo effect may emerge, let us consider the c block of the mean-field Hamiltonian and study the evolution of the electronic Fermi surfaces as a function of the filling n_c . By diagonalizing

$$\mathcal{H}_{\text{MF}}^c = -2t \sum_{\mathbf{k}\sigma} \begin{pmatrix} c_{\mathbf{k}A\sigma}^\dagger & c_{\mathbf{k}B\sigma}^\dagger \end{pmatrix} \begin{pmatrix} \cos k_1 & \cos k_2 + \cos k_3 \\ \cos k_2 + \cos k_3 & \cos k_1 \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}A\sigma} \\ c_{\mathbf{k}B\sigma} \end{pmatrix}, \quad (42)$$

we obtained the bare electronic spectrum

$$\epsilon_{\mathbf{k}\pm}^c = -2t [\cos k_1 \pm (\cos k_2 + \cos k_3)]. \quad (43)$$

Using the condition $n_c = \frac{1}{N} \sum_{i\alpha\sigma} \langle c_{i\alpha\sigma}^\dagger c_{i\alpha\sigma} \rangle$, we then solved for the chemical potential μ for various fillings n_c , and thus determined the Fermi surfaces shown in Fig. 4. It is clear that the system undergoes two Lifshitz transitions at $n_c \approx 0.45$ and $n_c = 1.5$. Coincidentally, these are the fillings at which the c Fermi surfaces intersect the Dirac cones of the Dirac spin liquid.

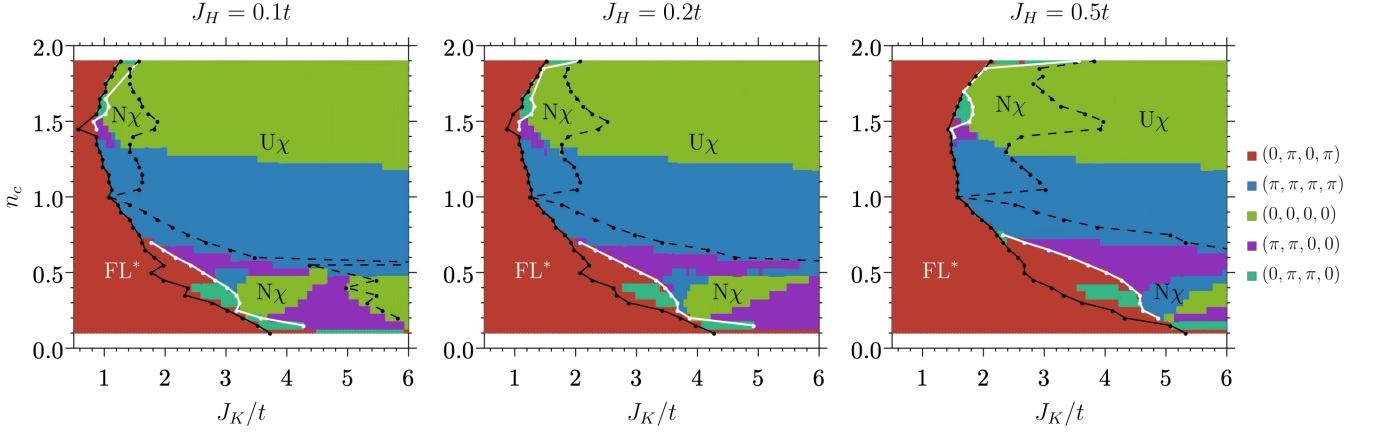


FIG. 5. Phase diagrams of the restricted mean-field theory for a system with $L_1 \times L_2$ unit cells and $L_1 = 2L_2 = 50$. We used fixed parameters $M = 2$, $T/t = 10^{-3}$, and $\kappa = 1$ for three different values of $J_H/t \in \{0.1, 0.2, 0.5\}$. The continuous black and white lines correspond to the right boundaries of the fractionalized Fermi liquid (FL*) and the regions with inhomogeneous b fields, respectively. The symbols $U\chi$ and $N\chi$ identify regions in which the absolute values of the χ fields are uniform (U) and nematic (N). The colors in the background correspond different flux sectors, as indicated by the legend.

D. Mean-field results for the restricted theory

Figure 5 depicts phase diagrams obtained from the numerical solution of the restricted mean-field theory (i.e., the theory with real χ variables). Notably, we find two regions with inhomogeneous hybridization fields: one above ($n_c > 1$) and another below ($n_c < 1$) half filling. Each such region includes at least two phases, which are distinguished by their flux configurations: $(0, \pi, 0, \pi)$ or $(0, \pi, \pi, 0)$. The former phase has the same flux configuration as the Dirac spin liquid and can be reached directly from the FL* via a *continuous* phase transition. **This raises the question: What is the mechanism driving the instability out of the FL* and toward a state with inhomogeneous Kondo screening?** A comparison between Figs. 5 and 4 suggests that this has something to do with the overlap between the bare c -electron Fermi surfaces and the Dirac points of the spin liquid.

Figure 6 gives a more detailed picture of the evolution of the mean-field parameters along two constant- n_c cuts through the $J_H = 0.2t$ phase diagram. The top row, corresponding to $n_c = 0.5$, only goes through the $(0, \pi, 0, \pi)$ inhomogeneous phase (red). The results show a clear splitting in the magnitudes of the hybridization fields, which vanish continuously as one approaches the FL* phase boundary. By contrast, the lower row applies to $n_c = 0.35$, for which we observe *two* intermediate inhomogeneous phases. Upon decreasing J_K , we see that one of the hybridization fields suddenly *vanishes*, whereas the remains nonzero. This type of sublattice-selective hybridization is characteristic of the $(0, \pi, \pi, 0)$ inhomogeneous phase (aquamarine). As J_K is decreased further, we find a first-order transition to a $(0, \pi, 0, \pi)$ inhomogeneous phase before the FL* is reached.

III. PROGRESS LOG

This is just meant to be an overview of how progress was made in this project.

- Initially, I considered a version of the mean-field theory with only three independent and real χ variables. These corresponded to $\chi_A = \chi_{A1}$, $\chi_B = \chi_{B1}$, and the intersublattice bond variables χ_{BA} , which were assumed to be equal to each other. I found that the ground state of the pure f sector was a nematic state with $|\chi_A| = |\chi_B| > |\chi_{BA}|$ and Dirac cones in its dispersion. The latter property was related to a flux structure with $\Phi_1 = \Phi_4 = 0$ and $\Phi_2 = \Phi_3 = \pi$. The solution of the full mean-field theory then revealed that, for certain fillings n_c , the increase of J_K could cause the system realize a phase with inhomogeneous Kondo screening upon exiting the unscreened phase. Further increase of J_K led to a two-step process in which the b fields would become uniform and, for larger J_K , so would the χ fields.
- Later, I relaxed the constraints on the χ fields to arrive at the version of the mean-field theory described in these notes. This generalization led to completely different solutions in the pure f sector. In the absence of a biquadratic interaction κ , I obtained a strongly dimerized state, which gives way to a gapped state with uniform

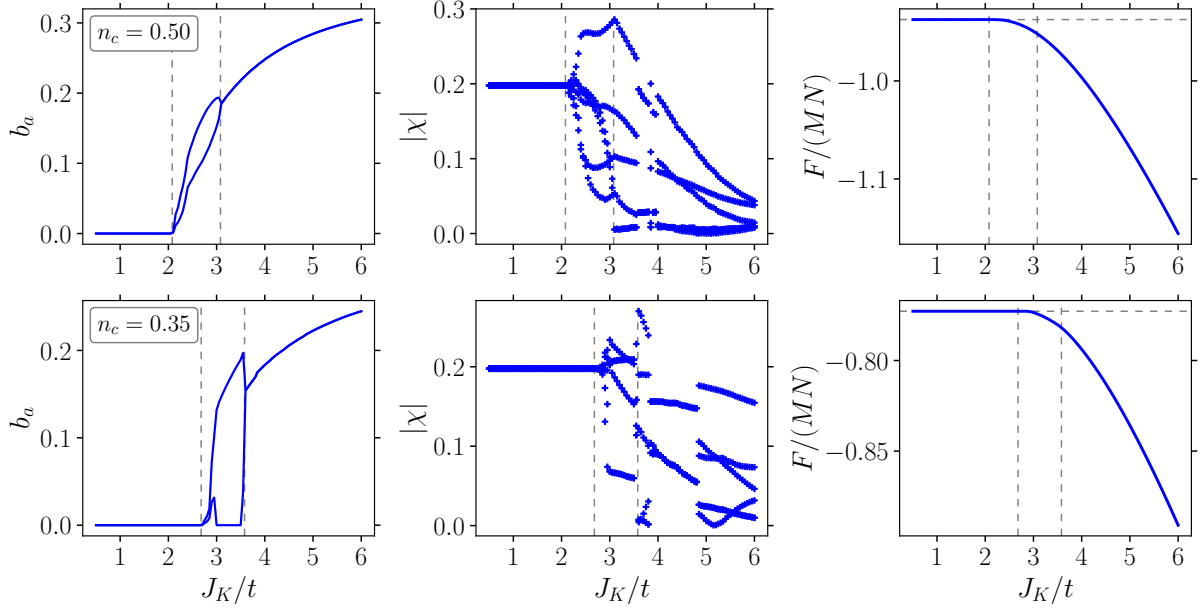


FIG. 6. Results along a cuts with constant $n_c = 0.5$ and 0.35 through the middle phase diagram ($J_H = 0.2t$) in Fig. 5. The vertical gridlines mark the transitions out of the FL* and Kondo hybridization wave phases.

$\pi/2$ fluxes and broken time-reversal symmetry above a critical value of κ . I was unable to reproduce a phase with inhomogeneous Kondo screening in this full unrestricted mean-field theory.

- I tried to stabilize a Dirac spin liquid in the pure f sector by adding a next-nearest-neighbor interaction J_2 . Unfortunately, this did not seem to work; hence the decision to restrict the mean-field theory to real values of χ . Notably, this restricted theory allowed for both the “isotropic” Dirac spin liquid described in the main text and the “nematic” Dirac spin liquid mentioned in the first bullet point. The latter did not appear as a viable solution. With this strategy, I was obtained a theory in which a Kondo hybridization wave emerges from an FL* state with a Dirac spin liquid through a continuous phase transition.

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